

Zihong Lu

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Education

Harbin Institute of Technology, Shenzhen	2024/09 – Present
M.Eng. in Control Science and Engineering, School of Intelligent Science and Engineering	Shenzhen, China
Harbin Institute of Technology, Shenzhen	2020/09 – 2024/06
B.Eng. in Automation, School of Intelligent Science and Engineering	Shenzhen, China

Research Interests

My research interests lie in autonomous aerial robotics, with a focus on vision-based navigation, active perception, reinforcement learning, and sim-to-real deployment for agile UAV systems.

Publications

- PerchRL: Vision-Based Agile Perching on Inclined Platforms under Rapid and Irregular Motion**
Zihong Lu, Zongzhuo Liu, Huaxu Li, Jinqiang Cui, Jie Mei, Youmin Gong, U Kei Cheang, Boyu Zhou
Under Review, CoRL 2026. [arXiv]
- Perception-aware Planning for Quadrotor Flight in Unknown and Feature-limited Environments**
Chenxin Yu, **Zihong Lu**, Jie Mei, Boyu Zhou
Accepted, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025). [arXiv]
- PUTN: A Plane-fitting based Uneven Terrain Navigation Framework**
Zhuzhu Jian*, **Zihong Lu***, Xiao Zhou, Bin Lan, Anxing Xiao, Xueqian Wang, Bin Liang
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2022). [arXiv]

Selected Research Projects

- STAR Group** 2024/09 – 2026/06
Project Lead, Vision-Based Agile Perching and Active Perception for Quadrotors; Advisor: Boyu Zhou
- Led an end-to-end reinforcement learning project for quadrotor perching on rapidly and irregularly moving inclined platforms, targeting robust landing under high speed (>2 m/s) and large tilt angles ($>70^\circ$).
 - Built Isaac Sim training environments, trained vision-guided landing policies, deployed sim-to-real experiments, and coordinated UAV platform integration and real-world flight tests.
 - Developed an active perception planning pipeline in AirSim + ROS, combining differential flatness, feature co-visibility, and MINCO trajectory parameterization for efficient navigation in unknown, feature-limited environments.
- MASLAB, Harbin Institute of Technology, Shenzhen** 2024/02 – 2024/06
Core Member, Vision-Closed-Loop Autonomous Drone Racing Through Gates
- Developed dynamic obstacle perception modules, handled ROS deployment and real-flight testing, and supported trajectory planning and system-level integration.
- Complex Terrain Robotics Group, Tsinghua Shenzhen International Graduate School** 2021/10 – 2022/03
Core Member / Co-first Author, Uneven Terrain Navigation for Ground Robots
- Participated in the design of the PUTN navigation framework; developed a plane-fitting-based PF-RRT* global planner, incorporated terrain traversability into planning costs, and designed an NMPC controller for real-time trajectory tracking.

Technical Skills

- **Programming Languages:** C/C++, Python, MATLAB
- **Robotics and Simulation:** Ubuntu, ROS1/ROS2, Gazebo, AirSim, Isaac Sim, UAV autonomous navigation, exploration, RL
- **Embedded Systems:** STM32, ESP32, embedded control and communication for mobile robot platforms

Selected Honors and Awards

Third Place - DJI RoboMaster University AI Challenge	Apr. 2024
National Silver Award - National College Students Engineering Practice and Innovation Ability Competition	Nov. 2021
National First Prize - TI Cup National Undergraduate Electronics Design Contest	Dec. 2021
National Second Prize - Lanqiao Cup Electronic Design Contest, MCU Design and Development	Jun. 2022
Excellent Creative Award, 2nd Nationally - Meituan Low-Altitude Economy Intelligent Flight Management Challenge	